# Intelligent Agent

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# Outline

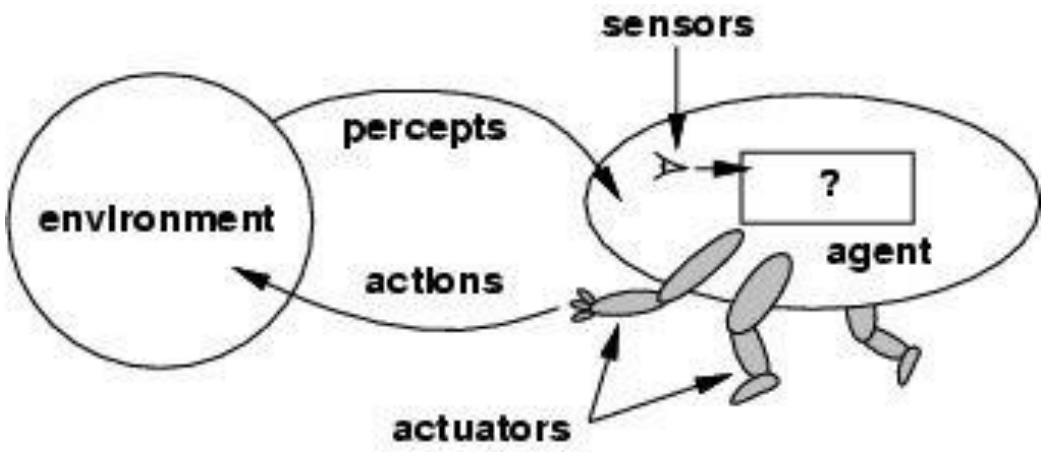
- Agents and environments
- PEAS (Performance measure, Environment, Actuators, Sensors)
- Rationality

# Intelligent Agent

- Artificial Intelligence focuses on building systems that can act intelligently in real-world environments.
- Examples:
  - Google Maps
  - Chatbot
  - Autonomous vehicle

# Intelligent Agents (continued)

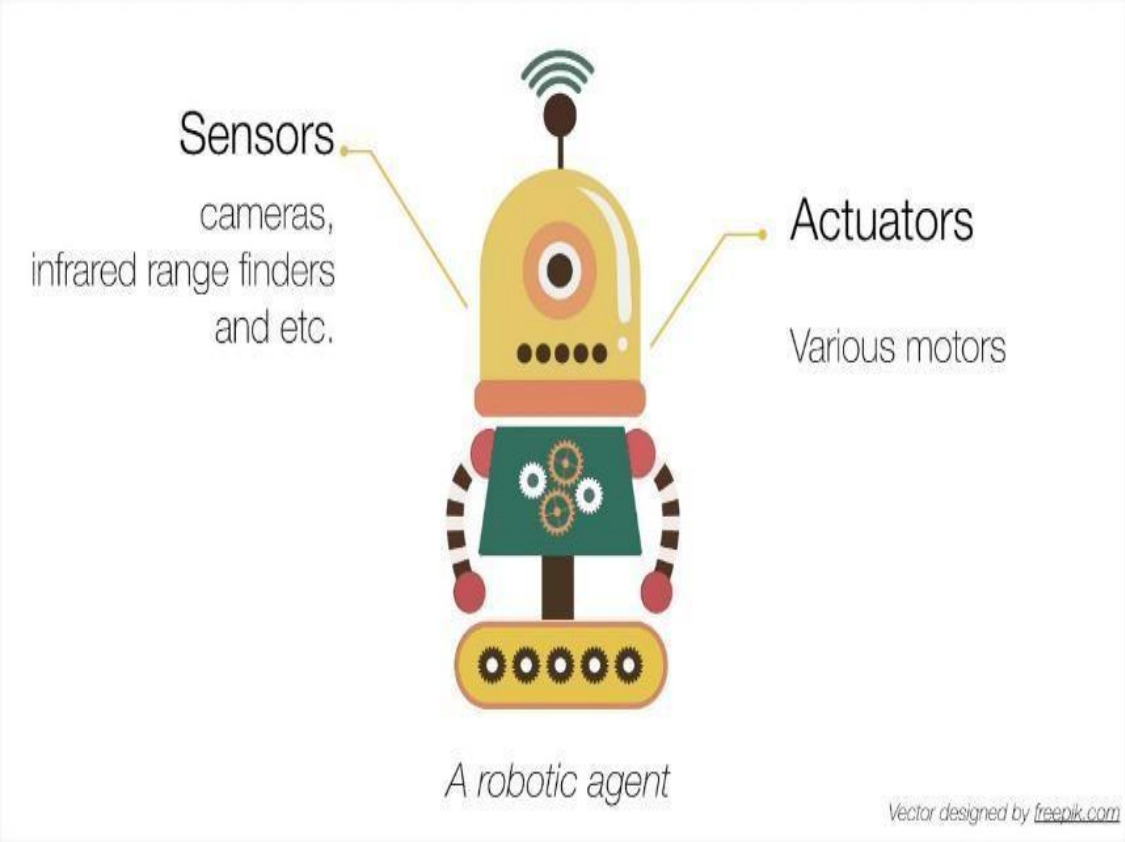
- An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.



# Intelligent Agents (continued)

- Sensor –
  - It is a device which detects the change in the environment and sends the information.
  - An agent observes it environment through sensors

- Actuators –
  - The component of machines that converts energy into motion.
  - The actuators are only responsible for moving and controlling a system.



# Example - Agents

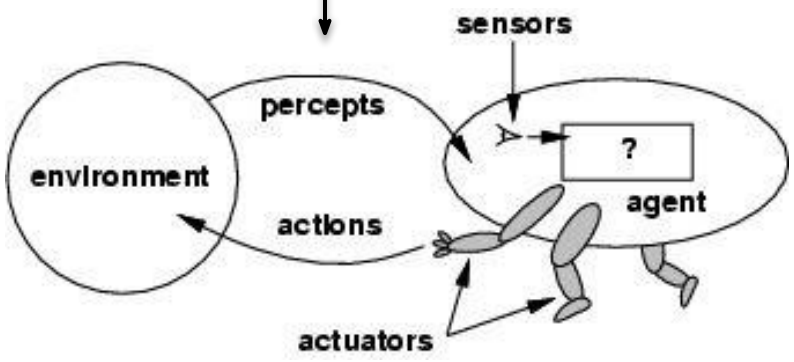
- Human agent:
  - eyes, ears, and other organs for sensors;
  - hands, legs,  mouth, and other body parts for actuators
- Robotic agent:
  - cameras and infrared range finders for sensors;
  - various motors for actuators
- Software agent:
  - Keystrokes, file contents as sensors
  - Act on those inputs and display output called as actuators

# Agent Terminology

- Percept
- Percept Sequence
- Behaviour of Agent
- Agent Function

# Agents & Environment

Input as : see/ hear etc... (percept sequence)




- The agent function maps from percept histories to actions:[$f$: $\mathcal{P}^{\star} \rightarrow \mathcal{A}$]
- The agent program runs on the physical architecture to produce $f$

agent = architecture + program

# Agent Function vs Agent Program

## Key Differences:

| Aspect | Agent Function | Agent Program |
|---|---|---|
| **Nature** | Abstract and mathematical | Concrete and implemented |
| **Purpose** | Idealized mapping of inputs to actions | Practical execution of decisions |
| **Flexibility** | Considers all possible inputs | Limited by computational and design constraints |
| **Real-World Use** | Guides theoretical understanding | Operates in real environments |

# PEAS

- It is a systematic way to define the properties of an intelligent agent.
- It helps in designing and analysing agents by breaking down their interaction with the environment.

**1.P**erformance Measure

**2.E**nvironment

**3.A**ctuators

**4.S**ensors

# Performance

- Behavior and performance of IAs in terms of agent function
    - Perception history (sequence) to Action Mapping
    - Ideal mapping: specifies which an agent should to take at any point in time.
- Performance measure: a subjective measure to characterize how successful an agent is (e.g., speed, power usage, accuracy, money, etc..)

# Environment

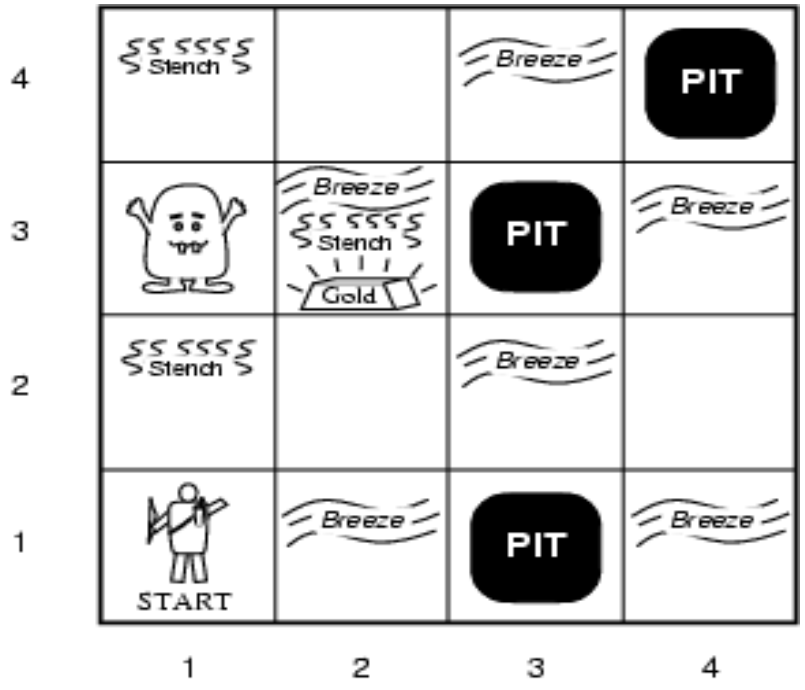
- Refers to the **external world** in which the agent operates.
- Includes everything the agent can perceive and interact with.

# Sensors & Effectors

- An Agent perceives its environment through sensors
  - The complete set of inputs at a given time is called a **percept**
  - The current percept, or a sequence of percepts can influence the actions of an agent
- It can change the environment through actuators / effectors
  - An operation involving an actuator is called an action
  - Action can be grouped into action Sequence

# PEAS – Wumpus World

- Performance measure
  - gold +1000, death -1000
  - -1 per step, -10 for using the arrow
- Environment
  - 4 X 4 grid of rooms
  - Agent always starts in square [1,1], facing to right
  - Squares adjacent to wumpus are smelly
  - Squares adjacent to pit are breezy
  - Glitter iff gold is in the same square
  - Shooting kills wumpus if you are facing it
  - Shooting uses up the only arrow
  - Grabbing picks up gold if in same square
  - Releasing drops the gold in same square




- Sensors:
  - Stench, Breeze, Glitter, Bump, Scream
- Actuators :
  - wheel/feet, arm

# PEAS - taxi driver

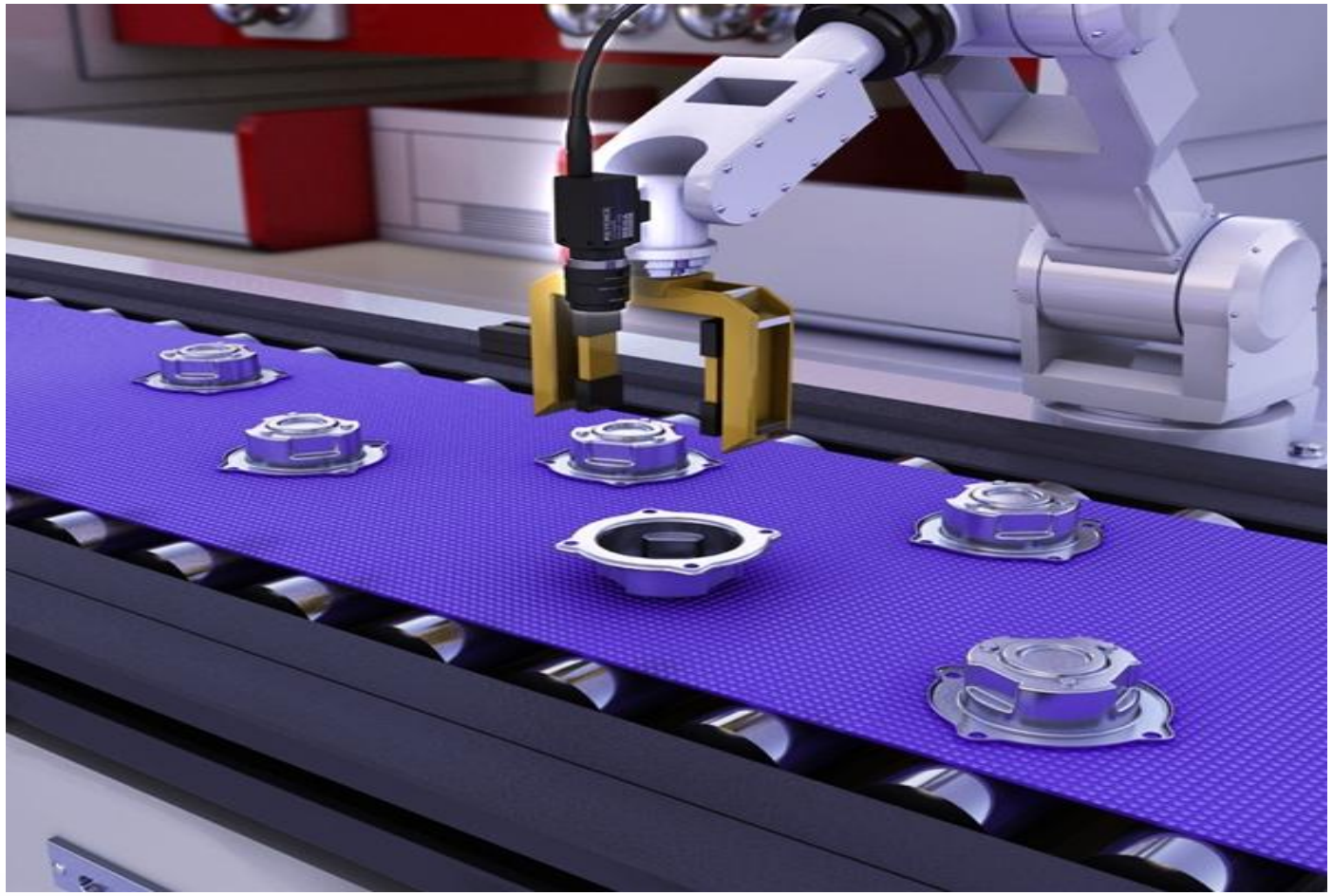
# PEAS - taxi driver (continued)

- Agent Type
  - Taxi Driver
- Performance Measure
  - Safe, fast, legal, comfortable trip, maximize profits
- Environment
  - Roads, other traffic, pedestrians, customers
- Actuators
  - Steering, accelerator, brake, signal, horn, display
- Sensors
  - Cameras, sonar, speedometer, GPS, odometer, engine sensors, keyboard

# Taxi – Driver Performance Measure

- Legal
  - Breaking signal ➔ -40 per signal break
  - Speeding ➔ -20 per speed limit violation
  - Wrong entry ➔ -30 for every wrong entry
  - Wrong parking => -20 for every wrong parking
  - Minor accident ➔ -70 for every minor accident
  - Major accident ➔ -100 for every major accident
- Safe
- Maximize profit
- Comfortable journey
  - Avoiding pothole ➔ +40 per pothole
  - Speeding at speedbreakers ➔ -30
  - Sudden break ➔ -30

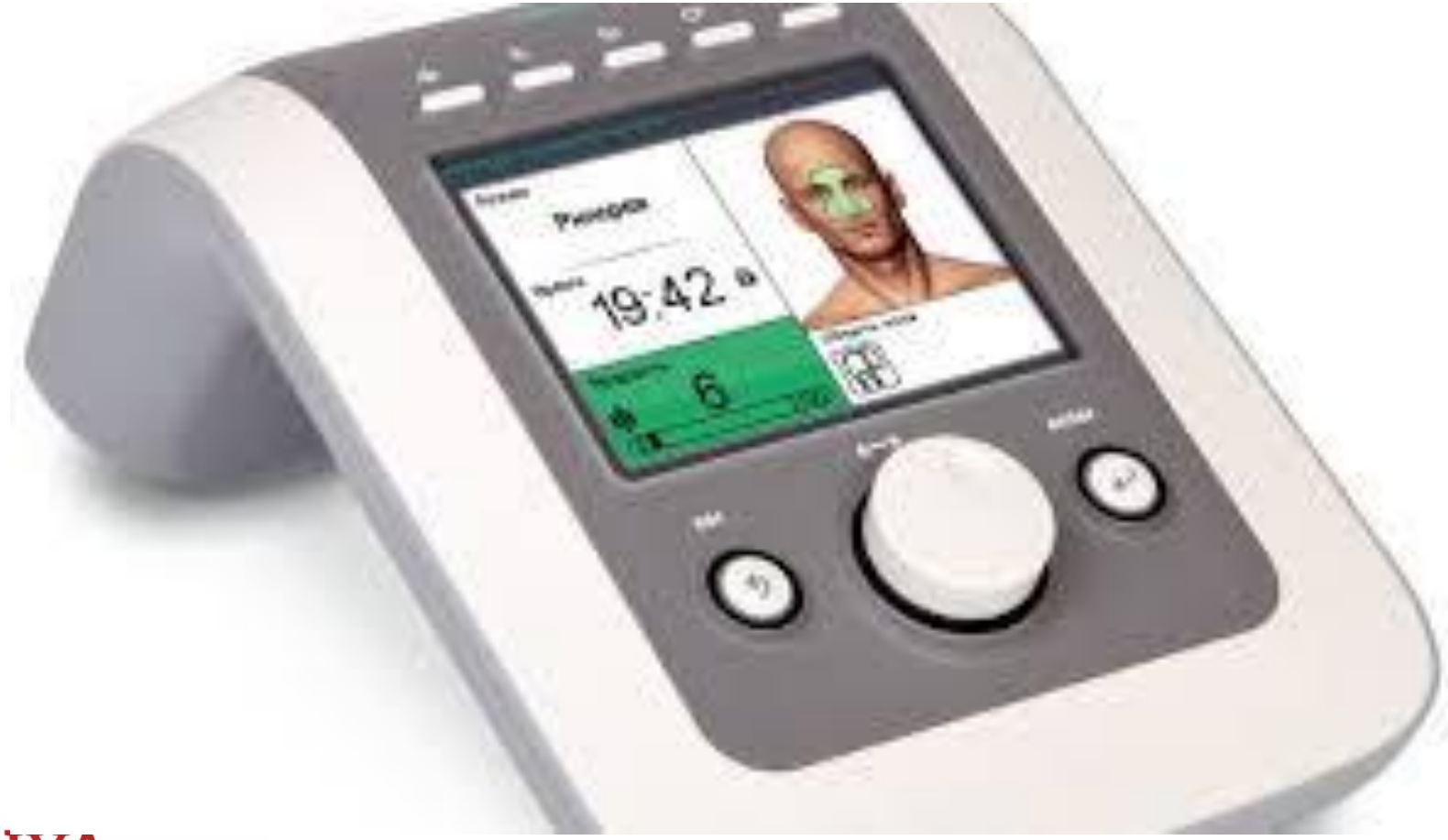
# Part-picking robot



# Part-picking robot (continued)

- Agent – Part-picking robot
- Performance measure
  - Percentage of parts in correct bins
- Environment
  - Conveyor belt with parts, bins
- Actuators
  - Jointed arm and hand
- Sensors
  - Camera, joint angle sensors

# Medical diagnosis



# Medical diagnosis (continued)

- Agent Type
  - Medical diagnosis
- Performance Measure
  - Healthy patient, reduced costs
- Environment
  - Patient, hospital, staff
- Actuators
  - Display of questions, tests, diagnoses, treatments, referrals
- Sensors
  - Keyboard entry of symptoms, findings, patient's answers

# Online Shopping Recommendation System

- **Performance Measure:**
- **Environment:**
- **Actuators:**
- **Sensors:**

# Smart Home Thermostat

- **Performance Measure:**
- **Environment:**
- **Actuators:**
- **Sensors:**

# Drone Delivery System

- **Performance Measure:**
- **Environment:**
- **Actuators:**
- **Sensors:**

# Functionalities of Rational Agent

- **Perceive:** Gather information from the environment through sensors.
- **Reason/Decide:** Process percepts to determine the best possible action.
- **Act:** Execute actions that influence the environment.
- **Learn:** Improve performance over time by adapting to new situations.
- **Achieve Goals:** Maximize success according to a performance measure (e.g., winning a game, cleaning a room efficiently).

# Components of Rational Agent

- **Sensors:** Input mechanisms to perceive the environment.
- **Actuators:** Output mechanisms to act on the environment.
- **Knowledge Base:** Stores facts, rules, and learned information.
- **Decision-Making Module:** Chooses rational actions based on percepts and goals.
- **Learning Module:** Updates knowledge and strategies from experience.

# Build Rational Agent

- **Define the Environment:** What the agent will operate in stochastic).
- **Specify Performance Measure:** What counts as success.
- **Design Architecture:** Combine sensors, actuators, and reasoning modules.
- **Implement Decision Logic:** Algorithms (rule-based, search, planning, learning).
- **Incorporate Learning:** Allow adaptation to improve over time.

# Rational Action

- The action that maximizes the expected value of the performance measure, given the percept sequence so far.
- **Rational = Best?** Yes, but only relative to the agent's knowledge.
- **Rational = Optimal?** Yes, but only within the agent's abilities and constraints.
  - Rationality is about *doing the best you can with what you know and the resources you have*.
  - Optimality is the theoretical best possible action, which may not be achievable in practice.

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# Intelligent Agent

Characteristics

- **Sense** → perceive the world.
- **Act** → influence the world.
- **Autonomous** → act independently.
- **Rational** → act logically to achieve goals.

# Question ?